

AI-BEFORE-MID — Weeks 1–8 · 17 Aug – 11 Oct 2026 ·

MIDTERM (W8)

Scope: Weeks 1–8 (17 Aug – 11 Oct 2026) — AI intro, uninformed search, informed search & A, CSPs & adversarial search, FOL & knowledge representation, genetic algorithms, ending at the Week 8 midterm. *Siblings:* [AI-NAV.md](#) (master map) · [AI-AFTER-MID.md](#) (Weeks 9–15) · [AI-LABS.md](#) (labs). Week-colored headers below match the NAV schedule. Exam split: ~40% numerical/trace + ~60% theory, weighted heavily toward the three pillars (A, GA, backprop). Process > final number on trace problems. **Exam order:** A*/search traces first, theory second.

At a glance (W1–W8)

W	Dates	Variant	File	Topics	Module
W1	17–23 Aug	P2	BEFORE	Intro to AI & problem solving	01
W2	24–30 Aug	P2	BEFORE	Uninformed search	02
W3	31 Aug–06 Sep	P2	BEFORE	Informed search & A*	02
W4	07–13 Sep	P0	BEFORE	CSPs & adversarial search	02
W5	14–20 Sep	P0	BEFORE	FOL & knowledge representation	03 + 05
W6	21–27 Sep	P2	BEFORE	Genetic algorithms	04
W7	28 Sep–04 Oct	P1	BEFORE	Midterm revision	01–06
W8	05–11 Oct	MIDTERM	BEFORE	Exam week — no new material	—

Tier key: **P0** = no exam pressure (front-load new material) · **P1** = light revision · **P2** = drill-heavy week. Variants editorial (AI manifests carry no variant field).

W1 — Intro to AI & Problem Solving · 17–23 Aug · P2 · 5 hrs

Banner: Rotation AI (Wed+Fri) · Tier P2 · Time budget 5 hrs · P0 floor: pack pass (narrative-sourced) + def/formula skim + same-problem drill.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 1	lines 18–33
Fear-Killer Pack	— (no pack — AI-W1 not in Fear-Killer-Packs.md)	—
Week manifest	weeks/AI-W1.md	full file
Chapter breakdown	— (no 03-Chapter-Breakdowns/ ; module M1)	—

Definitions (verbatim)

Term	Definition
Heuristic	A problem-solving technique that uses practical methods not guaranteed to be optimal.
A*	An informed search algorithm that uses $f(n) = g(n) + h(n)$, optimal if h is admissible.

Formulas (verbatim)

Formula	Statement
A* Search	$f(n) = g(n) + h(n)$

Type	Items
Diagrams	— (intro week; preview #2 BFS vs DFS)
Numericals	— (none; drill = 3 search-problem formulations)
Tricky	— (AI _{Top-10-Tricky-Concepts.md} deferred)
Top-100	— (AI _{Top-100-Questions.md} deferred)

Books · Chapters · Media

Resource	Where
Russell & Norvig	Ch.1–2 (AI, agent) · Ch.3 (search intro)
Gate Smashers	AI playlist videos 1–3

Fear-Killer Pack — none this week

No pack (AI-W1). Deep study from narrative + R&N Ch.1–2. W1 is the only "pure theory" week.

Narrative — Week 1 (verbatim)

Topics: What is AI?; Strong AI vs Weak AI; Turing Test; Problem types (deterministic, stochastic, adversarial); State space representation

Resources: Russell & Norvig Ch.1-2; Gate Smashers — AI (videos 1-3)

Practice: - Define AI in 4 ways (thinking humanly, acting humanly, thinking rationally, acting rationally) - Given a real-world problem: identify the state space, initial state, goal state, actions - 8-puzzle: model as a search problem (state = tile positions, action = move blank)

Lab: Lab 1 — Implementing Simple Neural Network Using Perceptron (logic gates); Python refresher + AIMA setup first

Deliverable: Search problem formulation for 3 real-world scenarios

Time budget: 5 hrs

Lab pointer

Lab this week: AI-LABS.md → **Lab 1 — Simple NN Using Perceptron** (map in Lab-Schedule.md).

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** narrative-sourced drill (define AI 4 ways · model 8-puzzle) · def/formula skim (Heuristic + A* preview) · same-problem drill (3 formulations)
- **Same-problem drill target:** 8-puzzle state/action/goal modeling < 5 min
- **Deliverable:** search problem formulation for 3 real-world scenarios
- **Trap:** don't over-invest — theory-only week; the labs (ANN) start here but run on their own timeline

W2 — Uninformed Search · 24–30 Aug · P2 · 5 hrs

Banner: Rotation AI (Wed+Fri) · Tier P2 · Time budget 5 hrs · P0 floor: pack pass (narrative-sourced) + def/formula skim + same-problem drill.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 2	lines 37–52
Fear-Killer Pack	— (no pack — AI-W2 not in Fear-Killer-Packs.md)	—
Week manifest	weeks/AI-W2.md	full file
Chapter breakdown	— (no 03-Chapter-Breakdowns/; module M2)	—

Definitions (verbatim)

Term	Definition
Heuristic	A problem-solving technique that uses practical methods not guaranteed to be optimal.

Formulas (verbatim)

Formula	Statement
—	(no uninformed-search formula in <code>Formula-Book.md</code> ; complexity bounds live in the narrative)

Diagrams · Numericals · Tricky · Top-100

Type	Items
Diagrams	<code>Diagram-Book.md</code> #2 BFS vs DFS traversal on sample graph
Numericals	— (none; trace BFS/DFS by hand)
Tricky	— (deferred)
Top-100	— (deferred)

Books · Chapters · Media

Resource	Where
Russell & Norvig	Ch.3 (search)
AIMA Python	search module — code for BFS/DFS/UCS

Fear-Killer Pack — none this week

No pack (AI-W2). Deep study from narrative + R&N Ch.3.

Narrative — Week 2 (verbatim)

Topics: BFS (breadth-first); DFS (depth-first); UCS (uniform cost search); Depth-limited search; Iterative deepening; Completeness, optimality, time and space complexity

Resources: Russell & Norvig Ch.3; AIMA Python code — search module

Practice: - For each algorithm: write the frontier data structure (queue, stack, priority queue) - Trace BFS and DFS on the same graph — note the different paths - Complexity analysis: BFS = $O(b^d)$ time and space; DFS = $O(b^m)$ time, $O(bm)$ space

Lab: Lab 2 — Developing an Artificial Neural Network (ANN) Using Perceptron (learning rule, fit/predict)

Deliverable: BFS and DFS implementation + trace output for a sample graph

Time budget: 5 hrs

Lab pointer

Lab this week: `AI-LABS.md` → **Lab 2 — ANN Using Perceptron** (map in `Lab-Schedule.md`).

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** narrative-sourced drill (frontier data structures; BFS/DFS trace) · def skim (Heuristic) · same-problem drill (complexity bounds)
- **Same-problem drill target:** BFS/DFS trace on a new graph < 10 min; complexity bounds from memory
- **Deliverable:** BFS and DFS implementation + trace output for a sample graph
- **Trap:** none flagged — foundation week for W3's A*

W3 — Informed Search & A* · 31 Aug–06 Sep · P2 · 6 hrs

Banner: Rotation AI (Wed+Fri) · Tier P2 · Time budget 6 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.

Sources & offsets

Source	Where	Ref
Narrative	<code>Week-by-Week-Narrative.md</code> → Week 3	lines 56–74

Fear-Killer Pack	<code>Fear-Killer-Packs.md</code> → AI-W3	lines 11–17
Week manifest	<code>weeks/AI-W3.md</code>	full file
Chapter breakdown	— (no <code>03-Chapter-Breakdowns/</code> ; module M2)	—

Definitions (verbatim)

Term	Definition
A*	An informed search algorithm that uses $f(n) = g(n) + h(n)$, optimal if h is admissible.
Admissible Heuristic	A heuristic that never overestimates the true cost to reach the goal.
Heuristic	A problem-solving technique that uses practical methods not guaranteed to be optimal.

Formulas (verbatim)

Formula	Statement
A* Search	$f(n) = g(n) + h(n)$
$g(n)$	actual cost from start to n
$h(n)$	estimated cost from n to goal
Optimality	Optimal if h is admissible (never overestimates)
Consistency	$h(n) \leq c(n, a, n') + h(n')$

Diagrams · Numericals · Tricky · Top-100

Type	Items
Diagrams	<code>Diagram-Book.md</code> #1 A* search trace (graph with OPEN/CLOSED lists)
Numericals	<code>Numerical-Book.md</code> #33 A trace · #34 OPEN/CLOSED list maintenance · #35* admissibility check
Tricky	— (deferred)
Top-100	— (deferred)

Books · Chapters · Media

Resource	Where
Russell & Norvig	Ch.3 (informed search)
Gate Smashers	A* search video

Fear-Killer Pack AI-W3 (verbatim)

Resources: Gate Smashers (Search videos) • R&N Ch.3 • AIMA Python Code

- Trace A* on a graph where the heuristic is NOT admissible. Show the OPEN/CLOSED lists step by step and demonstrate how non-admissibility causes a suboptimal solution to be returned first.
- Given the 8-puzzle initial state `[[2,8,3],[1,6,4],[7,0,5]]` and goal `[[1,2,3],[8,0,4],[7,6,5]]`, model as a search problem and trace A* with Manhattan distance heuristic. Show at least 3 expansions with the OPEN list priority queue.
- Compare BFS, UCS, and A *on a graph with varying edge costs*. Construct a 5-node graph where BFS finds a suboptimal path but A finds the optimal. Show the frontier at each step for all three algorithms.
- What is an admissible heuristic? Prove that if h is admissible, A* with graph search is optimal. Then construct a heuristic that is admissible but not consistent, and show how consistency could have pruned more nodes.

Narrative — Week 3 (verbatim)

Topics: Heuristic functions; Greedy best-first search; A search ($f = g + h$); Admissible and consistent heuristics; Optimality of A

Resources: Russell & Norvig Ch.3; Gate Smashers — A* search

Practice: - A* trace: show OPEN list, CLOSED list, f , g , h for each node at each step - Admissible heuristic: never overestimates the true cost - Consistent (monotonic) heuristic: $h(n) \leq c(n, a, n') + h(n')$ - Manhattan vs Euclidean distance for 8-puzzle

Killer trap: Students claim A* is always optimal. It is optimal ONLY if the heuristic is admissible. Non-admissible heuristics →

faster but not guaranteed optimal. This is a guaranteed exam question.

Lab: Lab 3 — Applying Data Preprocessing for ANN (imputation, scaling, one-hot, split)

Deliverable: A implementation + comparison table (BFS vs DFS vs A)

Time budget: 6 hrs

Lab pointer

Lab this week: AI-LABS.md → Lab 3 — Data Preprocessing for ANN (map in Lab-Schedule.md).

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q1 non-admissible trace + Q2 8-puzzle trace) · def/formula skim (A/Admissible/Heuristic + A formulas) · same-problem drill #33/#34
- **Same-problem drill target:** A* OPEN/CLOSED trace < 15 min; admissibility check < 30 s
- **Deliverable:** A implementation + comparison table (BFS vs DFS vs A)
- **Trap:** A* is optimal ONLY with an admissible heuristic — guaranteed exam question

W4 — CSPs & Adversarial Search · 07–13 Sep · P0 · 5 hrs

Banner: Rotation AI (Wed+Fri) · Tier P0 · Time budget 5 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 4	lines 78–96
Fear-Killer Pack	Fear-Killer-Packs.md → AI-W4	lines 19–25
Week manifest	weeks/AI-W4.md	full file
Chapter breakdown	— (no 03-Chapter-Breakdowns/; module M2)	—

Definitions (verbatim)

Term	Definition
CSP (Constraint Satisfaction Problem)	A problem where variables must be assigned values from domains subject to constraints.
Minimax	A decision rule for minimizing the maximum possible loss in adversarial search.

Formulas (verbatim)

Formula	Statement
Minimax — Value(state)	Utility(state) if terminal
Minimax — MAX value	max over successors of MIN-VALUE
Minimax — MIN value	min over successors of MAX-VALUE

Diagrams · Numericals · Tricky · Top-100

Type	Items
Diagrams	Diagram-Book.md #3 Minimax game tree with backed-up values · #4 Alpha-beta pruning on game tree
Numericals	— (none in Numerical-Book.md; hand-traces only)
Tricky	— (deferred)
Top-100	— (deferred)

Books · Chapters · Media

Resource	Where
Russell & Norvig	Ch.5 (adversarial) · Ch.6 (CSP)

Gate Smashers	CSP video · Minimax video
AIMA Python	csp.py (AC-3, backtracking, NQueensCSP) · games.py (minimax/alpha-beta)

Fear-Killer Pack AI-W4 (verbatim)

Resources: R&N Ch.5-6 • AIMA Python Code • UC Berkeley CS188 Pac-Man Projects

1. Formulate the N-Queens problem as a CSP. For N=8, trace backtracking search with forward checking. Count the number of assignments tried with and without forward checking. Where does forward checking prune search?
2. Apply AC-3 arc consistency to a CSP with variables A,B,C, domains {1,2,3}, and constraints: A < B, B < C, A ≠ C. Show the queue evolution step by step. Which values are eliminated and which arcs are re-added?
3. Trace Minimax on a 4-level game tree where leaf values are: 3,5,2,8,1,9,4,6,7,2,0,3,5,4,9,1. Show the backed-up values at each level. Then apply alpha-beta pruning on the same tree — mark which branches are pruned and explain why.
4. How does move ordering affect alpha-beta pruning efficiency? Given a tree where the optimal ordering visits O(b^(d/2)) nodes but worst-case visits O(b^d), construct a concrete 3-level example showing best vs worst ordering.

Narrative — Week 4 (verbatim)

Topics: Constraint Satisfaction Problems (CSPs); Backtracking search; Forward checking; Arc consistency (AC-3); Adversarial search; Minimax algorithm; Alpha-beta pruning

Resources: Russell & Norvig Ch.5-6; Gate Smashers — CSP, Minimax

Practice: - CSP formulation: variables, domains, constraints - Backtracking trace for N-Queens (4 or 8) - Minimax trace on a game tree — show backed-up values - Alpha-beta pruning: identify which branches are pruned, explain why ordering matters

Trap: Alpha-beta pruning effectiveness depends entirely on move ordering. Best moves first = maximum pruning. Worst moves first = no pruning.

Lab: Lab 4 — Developing ANN Using ADALINE (Widrow-Hoff, batch vs SGD)

Deliverable: Minimax trace + alpha-beta pruning example

Time budget: 5 hrs

Lab pointer

Lab this week: AI-LABS.md → Lab 4 — ANN Using ADALINE (map in Lab-Schedule.md).

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q2 AC-3 trace + Q3 minimax trace) · def/formula skim (CSP/Minimax + minimax formulas) · same-problem drill (N-Queens backtracking)
- **Same-problem drill target:** minimax backed-up values on a 4-level tree; alpha-beta prune decisions < 15 min
- **Deliverable:** Minimax trace + alpha-beta pruning example
- **Trap:** alpha-beta pruning depends entirely on move ordering — best moves first = max pruning

W5 — FOL & Knowledge Representation · 14–20 Sep · P0 · 5 hrs

Banner: Rotation AI (Wed+Fri) · Tier P0 · Time budget 5 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 5	lines 100–116
Fear-Killer Pack	Fear-Killer-Packs.md → AI-W5	lines 27–36
Week manifest	weeks/AI-W5.md	full file
Chapter breakdown	— (no 03-Chapter-Breakdowns/; modules M3 + M5)	—

Definitions (verbatim)

Term	Definition
FOL (First-Order Logic)	A formal logical system that uses quantifiers ($\forall$, $\exists$) and predicates to represent knowledge.

Resolution	A rule of inference for proving logical statements by refutation.
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Formulas (verbatim)

Formula	Statement
—	(no FOL formula in <code>Formula-Book.md</code> ; conversion/unification rules live in the narrative)

Diagrams · Numericals · Tricky · Top-100

Type	Items
Diagrams	<code>Diagram-Book.md</code> #14 Semantic network (nodes = concepts, edges = relations)
Numericals	— (none; 10 English-to-FOL conversions are the drill)
Tricky	— (deferred)
Top-100	— (deferred)

Books · Chapters · Media

Resource	Where
Russell & Norvig	Ch.8 (FOL) · Ch.9 (inference) · Ch.10 (knowledge representation)
Gate Smashers	FOL video

Fear-Killer Pack AI-W5 (verbatim)

Resources: Gate Smashers (FOL videos) • R&N Ch.8-9 (FOL), Ch.10 (KR) • AIMA Exercises

- Convert "Every student who takes AI and studies hard passes the exam. John is a student who takes AI but does not study hard. Therefore, John does not pass the exam." into FOL and prove by resolution refutation. Show the CNF conversion step by step and the resolution steps with substitutions.
- Find the Most General Unifier (MGU) for: $P(f(x, g(a, y)), h(z, b))$ and $P(f(g(b, w), z), h(x, c))$. Show the unification algorithm step by step. If no MGU exists, explain why.
- Given the knowledge base: $\forall x (Student(x) \wedge Studies(x) \rightarrow Passes(x))$, $Student(Alice)$, $Studies(Alice)$, show both forward chaining and backward chaining proofs. Count the inferences each method makes and compare their efficiency.
- Represent the following knowledge using a semantic network: "A dog is a mammal. Mammals have fur. Dogs have four legs. A poodle is a dog. Poodles are known for being intelligent. My pet Rex is a poodle." Then convert the semantic network into FOL axioms. What inferences can your network support?
- Design a frame-based representation for a library system. Define frames for Book, Member, and Loan with slots and facets. Show how inheritance works between a general "Item" frame and a "Book" subclass. How does your frame system handle default values and procedural attachments?
- Write a script for "eating at a restaurant" showing the entry conditions, roles, props, scenes (entering, ordering, eating, paying, leaving), and results. Now show how the script handles exceptions (e.g., the order is wrong or payment is declined).

Narrative — Week 5 (verbatim)

Topics: Propositional vs first-order logic; FOL syntax (constants, variables, predicates, functions, quantifiers); Semantic networks; Frames; Scripts

Resources: Russell & Norvig Ch.8-10; Gate Smashers — FOL

Practice: - Convert English sentences to FOL ($\forall, \exists, \wedge, \vee, \Rightarrow, \Leftrightarrow$) - Skolemization: remove existential quantifiers - Unification: find MGU (Most General Unifier) - Resolution refutation: prove a conclusion by contradiction

Deliverable: 10 English-to-FOL conversions + resolution proof

Lab: Lab 5 — Developing ANN Using Backward Propagation (multilayer + backprop on XOR)

Time budget: 5 hrs

Lab pointer

Lab this week: `AI-LABS.md` → **Lab 5 — ANN Using Backward Propagation** (map in `Lab-Schedule.md`).

- **P0 floor:** fear-killer pack pass (Q1 resolution refutation + Q2 MGU) · def skim (FOL/Resolution) · same-problem drill (English-to-FOL)
- **Same-problem drill target:** 5 English→FOL conversions < 20 min; resolution refutation step-by-step
- **Deliverable:** 10 English-to-FOL conversions + resolution proof
- **Trap:** none flagged — theory-heavy, self-contained week

W6 — Genetic Algorithms · 21–27 Sep · P2 · 6 hrs

Banner: Rotation AI (Wed+Fri) · Tier P2 · Time budget 6 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 6	lines 120–141
Fear-Killer Pack	Fear-Killer-Packs.md → AI-W6	lines 38–44
Week manifest	weeks/AI-W6.md	full file
Chapter breakdown	— (no 03-Chapter-Breakdowns/; module M4)	—

Definitions (verbatim)

Term	Definition
GA (Genetic Algorithm)	An evolutionary algorithm that uses selection, crossover, and mutation to optimize solutions.

Formulas (verbatim)

Formula	Statement
Selection Probability	$p(i) = f(i) / \sum f(j)$
Crossover	single-point, multi-point, uniform
Mutation	bit flip with probability p_m

Diagrams · Numericals · Tricky · Top-100

Type	Items
Diagrams	Diagram-Book.md #5 GA cycle (selection → crossover → mutation) · #6 Single-point crossover
Numericals	Numerical-Book.md #36 fitness calculation · #37 roulette wheel selection · #38 one generation trace
Tricky	— (deferred)
Top-100	— (deferred)

Books · Chapters · Media

Resource	Where
Russell & Norvig	Ch.4 (evolutionary computing)
Gate Smashers	GA video

Fear-Killer Pack AI-W6 (verbatim)

Resources: R&N Ch.4.1 · AIMA Python Code

1. Trace one complete GA generation for the knapsack problem with 6 items: weights [3,5,2,7,4,1], values [10,15,8,20,12,3], capacity = 10. Initial population of 4 chromosomes: [101100], [010111], [110010], [001101]. Show selection probabilities (roulette wheel), single-point crossover at position 3, and bit-flip mutation with $p=0.1$.
2. Explain the schema theorem and the building block hypothesis. Given a population where schemas $H1 = 1 ** 0 **$ and $H2 = 0 ** 1 **$ have different fitnesses, calculate the expected number of representatives in the next generation under proportionate selection.
3. Compare (μ, λ) vs $(\mu + \lambda)$ evolution strategies. For $\mu=5$ parents and $\lambda=20$ offspring, show how the next generation is selected

- in each case. Which strategy is more exploitative and which more explorative?
4. Design a GA for the traveling salesman problem with 8 cities. Show the encoding, crossover (PMX), and mutation (swap). Trace one generation showing fitness calculation based on total distance.

Narrative — Week 6 (verbatim)

Topics: Biological inspiration; GA cycle (selection → crossover → mutation); Roulette wheel selection; Tournament selection; Crossover (single-point, multi-point, uniform); Mutation; Fitness function; Evolutionary Strategies ((μ, λ) vs (μ + λ) selection, Rechenberg's 1/5 rule); Biology-Inspired Models (Particle Swarm Optimization, Ant Colony Optimization)

Resources: Russell & Norvig Ch.4 (evolutionary computing); Gate Smashers — GA

Practice: - Trace one complete GA generation by hand: 1. Initial population (bit strings) 2. Fitness computation 3. Selection probabilities (roulette wheel) 4. Crossover (single-point at a given index) 5. Mutation (flip a bit) - Compute selection probabilities from fitness values

Killer trap: Roulette wheel probability calculation: $p(i) = f(i) / \sum f(j)$. Students forget to divide by total fitness and give raw fitness values as probabilities. Examiners spot this immediately.

Lab: Lab 6 — Applying Uninformed Searching Techniques (BFS/DFS/UCS/IDS). *Note: GA is a theory pillar only — the workbook has no GA lab; self-study with aimacode/aima-python + Gate Smashers.*

Deliverable: GA trace (one full generation) + Python implementation

Time budget: 6 hrs

Lab pointer

Lab this week: AI-LABS.md → **Lab 6 — Uninformed Searching Techniques** (map in Lab-Schedule.md). *GA has no lab — theory pillar only.*

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q1 knapsack generation + Q4 TSP design) · def/formula skim (GA + selection/crossover/ mutation formulas) · same-problem drill #36/#37/#38
- **Same-problem drill target:** one full GA generation from memory; roulette-wheel $p(i) < 30$ s
- **Deliverable:** GA trace (one full generation) + Python implementation
- **Trap:** roulette wheel $p(i) = f(i)/\sum f(j)$ — don't hand raw fitness as probability (killer trap)

W7 — Midterm Revision · 28 Sep–04 Oct · P1 · 7 hrs

Banner: Rotation AI (Wed+Fri) · Tier P1 · Time budget 7 hrs · P0 floor: pack pass + def/formula skim + same-problem drill. **No new material.**

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 7	lines 145–160
Fear-Killer Pack	— (revision — no pack; review AI-W3/4/5/6)	—
Week manifest	weeks/AI-W7.md	full file
Chapter breakdown	— (all modules M1–M6)	—

Definitions (verbatim) — full W1–W6 sweep

Term	Definition
A*	An informed search algorithm that uses $f(n) = g(n) + h(n)$, optimal if h is admissible.
Admissible Heuristic	A heuristic that never overestimates the true cost to reach the goal.
Heuristic	A problem-solving technique that uses practical methods not guaranteed to be optimal.
CSP (Constraint Satisfaction Problem)	A problem where variables must be assigned values from domains subject to constraints.
Minimax	A decision rule for minimizing the maximum possible loss in adversarial search.
FOL (First-Order Logic)	A formal logical system that uses quantifiers ($\forall, \exists$) and predicates to represent knowledge.

Resolution	A rule of inference for proving logical statements by refutation.
GA (Genetic Algorithm)	An evolutionary algorithm that uses selection, crossover, and mutation to optimize solutions.

Formulas (verbatim) — full sweep

Formula	Statement
A* Search	$f(n) = g(n) + h(n)$
Consistency	$h(n) \leq c(n, a, n') + h(n')$
Minimax — Value(state)	Utility(state) if terminal
Selection Probability	$p(i) = f(i) / \sum f(j)$

Diagrams · Numericals · Tricky · Top-100

Type	Items
Diagrams	<code>Diagram-Book.md</code> #1–#6 full sweep (A* trace, BFS/DFS, minimax, alpha-beta, GA cycle, crossover)
Numericals	<code>Numerical-Book.md</code> #33–#38 (A*, OPEN/CLOSED, admissibility, fitness, roulette, generation)
Tricky	— (deferred)
Top-100	— (deferred)

Books · Chapters · Media

Resource	Where
Weeks 3–6 packs	timed past papers

Fear-Killer Pack — revision (AI-W3/4/5/6)

No single pack this week. Re-run the AI-W3/W4/W5/W6 packs under timed conditions.

Narrative — Week 7 (verbatim)

Topics: Comprehensive review of Weeks 1-6

Practice: - Timed past paper (2 hrs, closed book) - A* trace: one complete example with OPEN/CLOSED lists - GA generation: trace from scratch - FOL: 5 English-to-FOL conversions + resolution - Minimax: one game tree trace

Red flag: If you cannot trace A* on a simple graph within 15 minutes, repeat 3 daily until exam.

Lab: Lab 7 — Applying Informed Searching Techniques (A* + heuristic choice on 8-puzzle)

Sleep banking: Bedtime moves to 22:00 for 5 nights (Weeks 6-7). Sleep 9 hours. This protects memory consolidation of search algorithms, GA cycle, and FOL conversions during deep sleep.

Time budget: 7 hrs

Lab pointer

Lab this week: `AI-LABS.md` → **Lab 7 — Informed Searching Techniques** (map in `Lab-Schedule.md`).

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** re-run packs AI-W3/4/5/6 (timed) · def/formula skim (all W1–W6 defs) · same-problem drill #33/#37
- **Same-problem drill target:** A* trace < 15 min (repeat 3× daily if failing); GA generation from scratch
- **Deliverable:** timed past paper solved + A*/GA/FOL/minimax traces
- **Trap:** retrieval only — no passive re-reading; A* > 15 min → drill 3× daily until exam

W8 — MIDTERM EXAM WEEK · 05–11 Oct · Exam

Banner: Rotation AI (Wed+Fri) · Tier MIDTERM · No new deep study, no floor accrual. Ledger frozen during W8.

Sources

Source	Where	Ref
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Narrative	Week-by-Week-Narrative.md → Week 8	lines 164–174
Pack	Fear-Killer-Packs.md → Week 8	— (no pack)
Week manifest	weeks/AI-W8.md	full file

Fear-Killer Pack — Week 8 (verbatim)

No new pack. Active recall only — review the 1-page cheat sheet and sleep 8 hours.

Narrative — Week 8 (verbatim)

Focus: Active recall only. No new material.

- Review search algorithm comparison table (BFS, DFS, UCS, A*)
- Review GA cycle (selection → crossover → mutation)
- Review FOL conversion rules
- Sleep 8 hours each night

Exam-day stack (Wed + Fri)

- [] Past-paper run for the exam subject — 60 min, P0
- [] Blank-page retrieval of that subject — 30 min, P0
- [] Master Error Log review — 20 min, P0

Notes

- Review search algorithm comparison table; GA cycle; FOL conversion rules. Sleep 8 h each night.
- Exam order: *A/search traces first, theory second*. A trace target < 15 min.
- **Ledger frozen during Wk8** — no new accrual; cleared in Wk9 recovery.